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$$A(z) = z^3 - z^2 - z - 2$$

$$\begin{array}{c|cccc} & 1 & -1 & -1 & -2 \\ 2 & & 2 & 2 & 2 \\ \hline & 1 & 1 & 1 & 0 \end{array}$$

$$= (z - 2)(z^2 + z + 1)$$

$$D = 1^2 - 4 \cdot 1 \cdot 1 = -3$$

$$\sqrt{D} = \pm\sqrt{3}i$$

$$\Rightarrow z_1 = \frac{-1 + \sqrt{3}i}{2}$$

$$\Rightarrow z_2 = \frac{-1 - \sqrt{3}i}{2}$$

$$A(z) = (z - 2) \left( z + \frac{1 + \sqrt{3}i}{2} \right) \left( z + \frac{1 - \sqrt{3}i}{2} \right)$$

## References